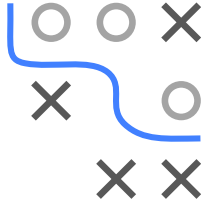
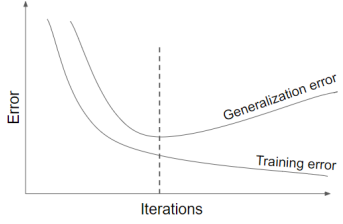
Introduction to Machine Learning



Regularization
Early Stopping

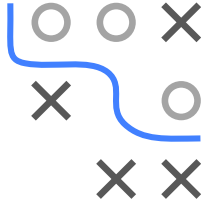


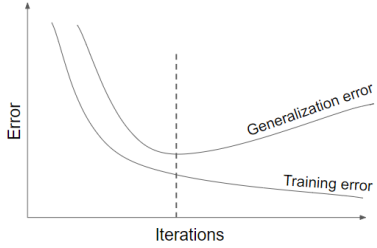
Learning goals

- Know how early stopping works
- Understand how early stopping acts as a regularizer

EARLY STOPPING

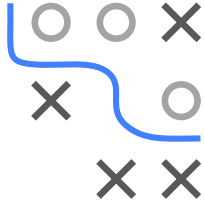
- Especially for complex nonlinear models we can easily overfit
- In optimization: Often, after a certain number of iterations, generalization error begins to increase even though training error continues to decrease





EARLY STOPPING AND *L2* ▸ Goodfellow, Bengio, and Courville 2016

Strengths	Weaknesses
Effective and simple	Periodical evaluation of validation error
Applicable to almost any model without adjustment	Temporary copy of $\boldsymbol{\theta}$ (we have to save the whole model each time validation error improves)
Combinable with other regularization methods	Less data for training $\rightarrow$ include $\mathcal{D}_{\text{val}}$ afterwards



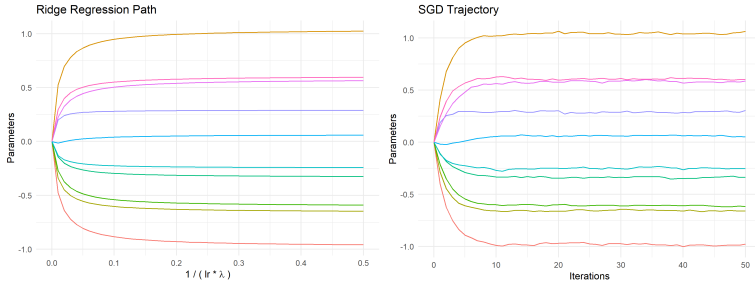
- For simple case of LM with squared loss and GD optim initialized at $\boldsymbol{\theta} = 0$: Early stopping has exact correspondence with *L2* regularization/WD: optimal early-stopping iter T_{stop} inversely proportional to λ scaled by step-size α

$$T_{\text{stop}} \approx \frac{1}{\alpha\lambda} \Leftrightarrow \lambda \approx \frac{1}{T_{\text{stop}}\alpha}$$

- Small λ (regu. $\downarrow$) $\Rightarrow$ large T_{stop} (complexity $\uparrow$) and vice versa

SGD TRAJECTORY AND *L2* ▸ Ali, Dobriban, and Tibshirani 2020

Solution paths for *L2* regularized linear model closely matches SGD trajectory of unregularized LM initialized at $\theta = 0$



Caveat: Initialization at the origin is crucial for this equivalence to hold, which is almost never exactly used in practice in ML/DL applications